

Decision Tree

Algorithm

Classification & Regression — Learns by asking questions

- R.M.K. Engineering College
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Topics

- What is a Decision Tree?
- Key terminology
- How it learns
- Gini Impurity
- Information Gain
- Worked example
- Overfitting & Pruning
- Pros, Cons & Uses

What is a Decision Tree?

Think of a doctor who asks questions to diagnose a patient: 'Do you have fever?' → Yes → 'Do you have cough?' → Yes → 'Flu' — that is exactly how a Decision Tree works!

Tree structure

A hierarchical model made of nodes and branches — like an upside-down tree

Ask questions

Each internal node asks a question about one feature of the data

Reach a decision

Each leaf node gives the final answer — a class label or a value

Classification Tree

Output is a class label

Example: Play cricket? Yes / No

Example: Fruit → Mango / Apple / Banana

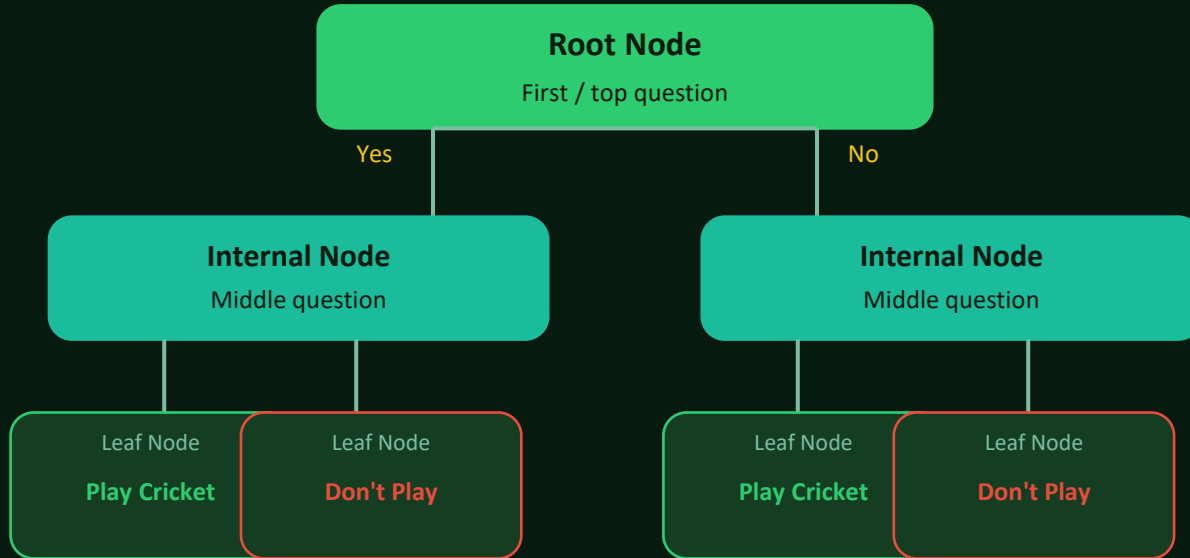
Regression Tree

Output is a continuous number

Example: Predict house price

Example: Predict student score

Parts of a Decision Tree — Vocabulary



How a Decision Tree Learns — Step by Step

1

Start with all training data

All samples are at the root node in one mixed group — impure

2

Try every feature & threshold

For each feature, try all possible split points (e.g. Age < 30, Age < 45...)

3

Calculate Gini Impurity / Info Gain

Measure how pure each resulting split is — pick the best one

4

Split the data

Divide data into two child nodes using the best feature and threshold

5

Repeat recursively

Each child node now becomes a new root — repeat steps 2 to 4 on it

6

Stop when pure or limit reached

Stop when: all same class, max depth reached, or too few samples left

Gini Impurity — How to Choose the Best Split

Gini Impurity Formula

$$\text{Gini}(\text{node}) = 1 - \sum p_i^2$$

where p_i = proportion of class i in the node
Information Gain = Gini(parent) - Weighted average
Gini(children)

Pure node (best)

All 5 samples: Class A



$$1 - (1.0^2 + 0.0^2) = 0.00$$

Perfect — no mixing

Mixed node

3 Class A, 2 Class B



$$1 - (0.6^2 + 0.4^2) = 0.48$$

Moderate mixing

Equally mixed (worst)

5 Class A, 5 Class B



$$1 - (0.5^2 + 0.5^2) = 0.50$$

Maximum impurity

Lower Gini = Purer node = Better split | Always pick the split with HIGHEST Information Gain

Choosing the Best Split — Step by Step Calculation

Parent node (10 samples)

6 Play Cricket + 4 Don't Play

$$\text{Gini} = 1 - (0.6^2 + 0.4^2) = 0.48$$

Split question: Is outlook sunny?

Left child — Sunny = Yes

Samples: 5

4 Play + 1 Don't Play

$$\text{Gini} = 1 - (0.8^2 + 0.2^2) = 0.32$$

Right child — Sunny = No

Samples: 5

2 Play + 3 Don't Play

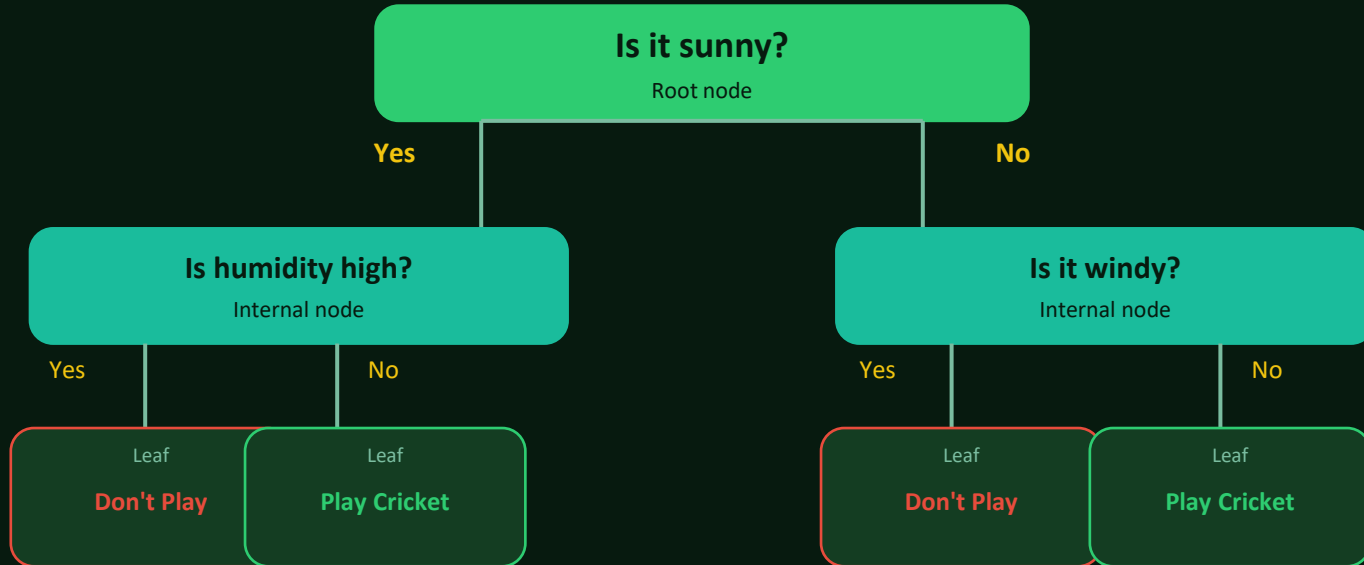
$$\text{Gini} = 1 - (0.4^2 + 0.6^2) = 0.48$$

Information Gain Calculation

$$\text{IG} = 0.48 - [(5/10 \times 0.32) + (5/10 \times 0.48)] = 0.48 - 0.40 = 0.08$$

Try this for all features — pick the feature with the HIGHEST Information Gain as the split

Decision Tree: Should We Play Cricket Today?



Predict for: Sunny=Yes, Humidity=Normal, Windy=No

Sunny? → Yes → Humidity high? → No → Leaf node → PLAY CRICKET

Gini Impurity vs Entropy vs Variance Reduction

Gini Impurity

$$\text{Gini} = 1 - \sum p_i^2$$

Range:

0 (pure) to 0.5 (max)

Used for:

Classification tasks

Algorithm:
m:

CART algorithm
(scikit-learn default)

Speed:

Yes —
computationally
faster

Entropy / Information Gain

$$H = -\sum p_i \log_2(p_i)$$

Range:

0 (pure) to 1 (max for
binary)

Used for:

Classification tasks

Algorithm:
m:

ID3, C4.5 algorithms

Speed:

Slower — uses
logarithm

Variance Reduction

$$\text{Var} = \sum (y_i - \bar{y})^2 / n$$

Range:

0 = best split

Used for:

Regression trees only

Algorithm:
m:

CART for regression

Speed:

Yes

For exams: Gini Impurity is most commonly tested — lower Gini = better split

The Overfitting Problem & How Pruning Solves It

The Problem — Overfitting

A very deep tree memorizes every training sample perfectly — but fails badly on new data. It learns the 'noise' instead of the pattern.

Shallow tree (depth 2)

Underfit

Underfits — too simple, misses patterns, low training accuracy

Optimal tree (depth 5)

Just right

Good balance — learns real patterns, high test accuracy

Deep tree (depth 20)

Overfit

Overfits — memorizes noise, 100% train accuracy but poor test accuracy

Pre-pruning (Early stopping)

Stop growing when: max_depth reached, min_samples_split not met, Gini improvement too small

Post-pruning

Grow the full tree first, then remove branches that don't improve accuracy on validation data

How We Evaluate a Decision Tree Model

Actual vs Predicted	Predicted: Positive	Predicted: Negative
Actual: Positive	TP = 40 Correctly identified	FN = 8 Missed positives
Actual: Negative	FP = 5 False alarms	TN = 47 Correctly identified

Accuracy	$(TP+TN) / \text{Total}$
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Precision	$TP / (TP+FP)$
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Recall	$TP / (TP+FN)$
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F1 Score	$2 \times P \times R / (P+R)$
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Decision Tree — Strengths, Weaknesses & Real-World Uses

Advantages

- ✓ Easy to visualise and explain
- ✓ No feature scaling needed
- ✓ Handles numbers & categories
- ✓ Shows feature importance clearly
- ✓ Fast prediction on new data
- ✓ Works for classification AND regression

Limitations

- ✗ Prone to overfitting (deep trees)
- ✗ Sensitive to small data changes
- ✗ Biased toward high-cardinality features
- ✗ Not good for complex linear patterns
- ✗ Unstable — small change, big tree change
- ✗ Solution: Random Forest (ensemble)

M

Medical

Diagnosis

F

Finance

Credit risk

E

Education

Pass/Fail

R

Retail

Churn predict

A

Agriculture

Crop disease

Summary — Decision Tree Algorithm

1

A Decision Tree is a flowchart of yes/no questions — each node splits data, each leaf gives the answer

2

It chooses each split using Gini Impurity or Entropy — always picks the split with highest Information Gain

3

Training = recursive splitting from root → children until data is pure or stopping condition is met

4

$Gini = 1 - \sum p_i^2$ | Range: 0 (pure) to 0.5 (max mix) | Lower Gini = better split

5

Overfitting is the main risk — controlled by max_depth, min_samples, and pruning techniques

6

Evaluate using Accuracy, Precision, Recall, F1. For complex tasks → use Random Forest (many trees)

Key formula: $Gini = 1 - \sum p_i^2$ | Info Gain = $Gini(\text{parent}) - \text{Weighted Gini}(\text{children})$